

1. Description of the Solution:

The solution is a HLSM with two processes and three states. The idle state is where the reset signal takes you, and the machine remains there until there is a start signal. Then, the machine jumps to the 'First Read' state where it enables the RAM and sends the first address to read. Subsequently, the machine goes into the 'reading' state where it stores the first value and then compares the rest of the entries in the memory to this one. Once the whole memory has been checked the control unit goes back to idle state. This behavior is shown in the state diagram in Figure 1. The register of the first entry is stored in the signal '*firstentry*'. The count of the appearances of the first entry is stored in the register '*count*' which is updated every clock cycle through the register '*newcount*'. The same happens with registers '*currentstate*' and '*newstate*' and with '*curraddress*' and '*newaddress*' since this is the standard way to handle registers in a two process HLSM description.

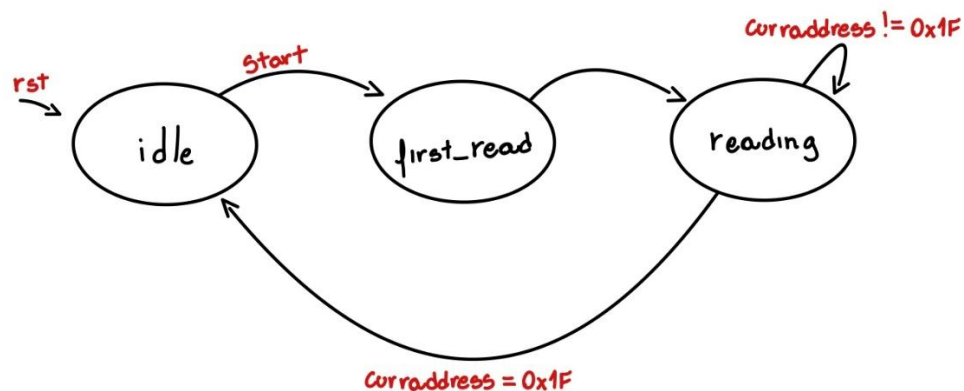


Figure 1: State Diagram

2. List of Files:

- control_unit.vhd: The main solution of the assignment. It describes the implementation fulfilling the requirements of the task.
- Testbench.vhd: This is the testbench developed to simulate the behavior of the control unit connected to the RAM.
- RAM.vhd: This is the file provided that was imported and did not suffer any changes.
- memory.mem: This file contains the memory initialization. I was filled with random 8-bit binary values from entry 0 to 31. The random values were generated manually. Eight values equal to the first entry were placed among the data in the file. This is value 30 in decimal or 0x1E in hexadecimal.
- Testbench_behav.wcfg: Wave configuration file with the main signals to verify the behavior of the state machine.

3. Simulation Results:

The simulation was carried out using the testbench developed for this purpose. The connection between the RAM and the control unit were also made in this description. The Testbench starts by resetting the machine, then the 'start' signal is activated. After the machine finishes reading all addresses. This part of the test verifies the correct behavior of the machine under normal conditions and it correctly outputs the number of appearances of the first entry. In Figure 2, it is possible to see how the machine behaves according to the specifications of the problem. The signal 'RAMDout' is the data that is outputted by the memory. The signal 'output' is a 5-bit output showing the result of the count of entries that are equal to the first entry. 'ADD' is the signal that the control unit sends to the memory and contains the address of the value to read. Additionally, the signals 'currentstate', 'firstentry', 'count' and 'curraddress' were included in the simulation for further comprehension of the description.

To test the reset process and the robustness of the machine after the end of its normal behavior, it is started again and interrupted with the reset signal. In Figure 3, the machine starts again in the clock cycle following the time stamp of 357.00 ns, until it finishes its process successfully in timestamp 560.00 ns showing the same result as the previous test.

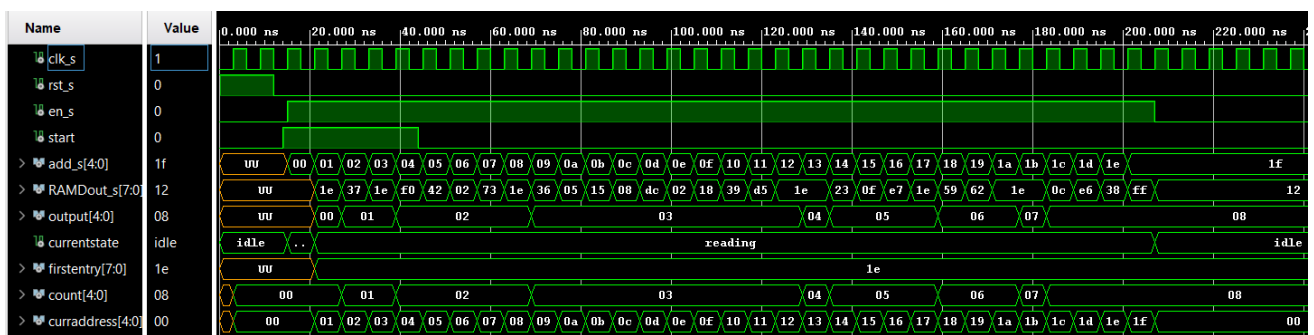


Figure 2: Simulation in Normal Conditions

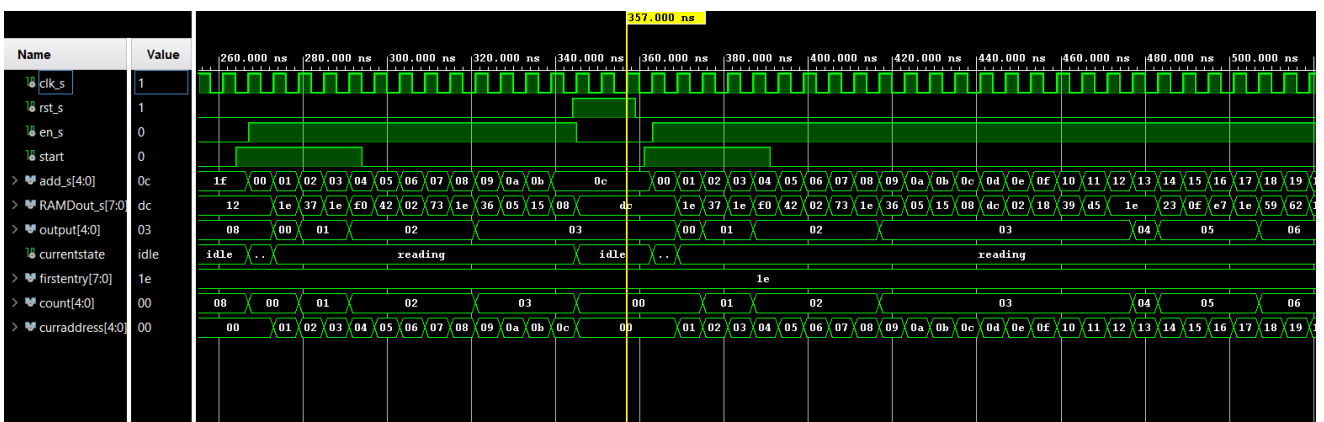


Figure 3: Reset and Restart Test